Longest Run of Identical Characters

Problem statement

Given a non-empty lowercase English string S, find the length of its longest contiguous block of equal characters.

Input format

One lowercase string S (no spaces) on one line.

Output format

Print one integer: the longest equal-character run length.

Constraints

- 1 <= length(S) <= 1000
- Every character is a-z

Sample input and output

Sample input	Expected output
`aaabbccccdd`	4

Important boundary conditions

A one-character word has answer 1. Runs need not appear at the beginning.

Implementation expectations

O(L) time and O(1) extra space are possible.

Submission requirements

Submit exactly one .c file. Use stdin and stdout without interactive prompts. Compile with GCC in C11 mode. Avoid undefined behavior, release dynamically allocated memory, and print precisely the required output. Follow the stated input constraints.

Teacher resources (not part of student handout)

A reference oracle, five sample student submissions, assignment-specific rules.json, and 12 verified tests are supplied separately in the instructor test pack. Do not distribute the reference oracle or deliberately faulty variants as student solutions.